

Alexa Mousley

alexa.mousley@mrc-cbu.cam.ac.uk
[Google Scholar](#), [Website](#), [GitHub](#)

EDUCATION

2021 – 2025	PhD	Medical Sciences	University of Cambridge, UK
2016 – 2020	BA	Neuroscience and Behaviour (with honours)	Vassar College, USA

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

Mousley A., He, Y., Vedeckina, M., Astle, D.E. (in prep). Wired for resilience: shared mental health and genetic structural connectivity patterns encode effects of adversity. Will submit to *PNAS*.

Poli, F., Fakhar, K., **Mousley, A.**, Klein, C., Wiesmann, C., Astle, D.E. (in prep). Curiosity shapes brain-like architectures and functions. Submitted to *Science*.

Dendorfer, A., Luppi, A., Poli, F., **Mousley, A.**, Astle, D., Fakhar, K. (in prep). Comprehensive benchmarking of distance metrics for generative connectome models. Will submit to *Network Neuroscience*.

Jelen, B. M., **Mousley, A.**,...& Astle, D. (in prep). The topology of adolescent mental health. Will submit to *Nature Mental Health*.

Trachtenberg, E., **Mousley, A.**, Jelen, M., & Astle, D.E. (2026). Mapping social profiles in childhood and adolescence: associations with cognition and brain structure. *bioRxiv*. <https://doi.org/10.64898/2026.04.20.719698>

Mousley, A. (2025). Formation, refinement, and reorganisation of complex brain network topology across the lifespan [Apollo - University of Cambridge Repository]. <https://doi.org/10.17863/CAM.121991>

Mousley, A., Bethlehem, R., Yeh, F-C., & Astle, D.E. (2025). Topological turning points across the human lifespan. *Nature Communications*, 16(1), 10005, <https://doi.org/10.1038/s41467-025-65974-8>

Poli, F., Oldham, S., **Mousley, A.**, Bullmore, E.T., Vértés, P.E., & Astle, D.E. (2025). Right time, right place: Heterochronicity shapes brain network formation. *bioRxiv*. <https://doi.org/10.1101/2025.10.13.682136>

Mousley, A., Akarca, D., & Astle, D. E. (2025). Premature birth changes wiring constraints in neonatal structural brain networks. *Nature Communications*, 16(1), 490. <https://doi.org/10.1038/s41467-024-55178-x>

Mitchell, D.J., **Mousley A.L.S.**, Shafto, M.A., & Duncan, J. (2023). Neural contributions to reduced fluid intelligence across the adult lifespan. *Journal of Neuroscience*, 43(2), 293-307. <https://doi.org/10.1523/JNEUROSCI.0148-22.2022>

Lawson, K.,...**Mousley, A.L.**,...& Bergstrom, H.C. (2022). Adolescence alcohol exposure impairs extinction and alters medial prefrontal cortex plasticity. *Neuropharmacology*, 21, 109048. <https://doi.org/10.1016/j.neuropharm.2022.109048>

Geary, C.G.,...**Mousley, A.L.**,...& Bergstrom, H. (2021). Sex differences in gut microbiota modulation of aversive conditioning, open field activity, and basolateral amygdala dendritic spine density. *Journal of Neuroscience Research*, 99(7), 1780-1801. <https://doi-org.ezp.lib.cam.ac.uk/10.1002/jnr.24848>

Scarlata, M.J.,...**Mousley, A.**,...& Bergstrom, H.C. (2019). Chemogenetic stimulation of the infralimbic cortex reverses alcohol-induced fear memory. *Scientific Reports*, 9(1), 1-15. <https://doi.org/10.1038/s41598-019-43159-w>

POSTERS & CONFERENCES

***Mousley, A.**, He, Y., Vedeckina, M., Astle, D.E. (Upcoming: 2026, August). A data-driven approach for identifying multi-domain connectivity signatures. Poster at *Cognitive Computational Neuroscience*, New York, USA.

***Mousley, A.**, Bethlehem, R., Yeh, F-C., & Astle, D. (2025, September). Turning points across the human lifespan. Poster at *FLUX*, Dublin, Ireland.

***Mousley, A.**, Bethlehem, R., Yeh, F-C., & Astle, D. (2025, July). Turning points across the human lifespan. Poster at *UK Neural Computing Conference*, Imperial College, London, UK.

***Mousley, A.**, Bethlehem, R., Yeh, F-C., & Astle, D. (2025, June). Turning points across the human lifespan. Poster at *Cambridge Imaging Festival*, University of Cambridge, Cambridge, UK.

***Mousley, A.**, Bethlehem, R., & Astle, D. (2024, August). Turning points across the human lifespan. Poster at *Cognitive Computational Neuroscience*, Massachusetts Institute of Technology, Cambridge, MA, USA.

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***Mousley, A.**, Akarca, D., & Astle, D. (2023, July). Premature birth changes wiring constraints in neonatal structural brain networks. Poster at *Organization for Human Brain Mapping*, Montreal, Canada.

***Mousley, A.**, Akarca, D., Carozza, S., & Astle, D. (2022, September). A generative network model of neonatal brain development. Poster at *Cambridge Neuroscience Seminar*, Cambridge, UK.

Wilk, V.,...***Mousley, A.L.**,...& Bergstrom, H.C. (2019, October). Sex dependent influence of gut dysbiosis on defensive behaviors and basolateral amygdala dendritic morphology. Poster at *Society for Neuroscience*, Chicago, IL.

* Indicates lead presenter.

TALKS & PRESENTATIONS

***Mousley, A.** (*Upcoming*: 2026, September). Adolescent Brain Matures at 32. 10th International Workshop on Adolescence, SRHR & HIV 2026, Bangkok, Thailand.

***Mousley, A.** (2026, June). Lifespan Connectomics Meets AI: Complexity, Development and Clinical Applications Symposium. Organization for Human Brain Mapping conference, Bordeaux, France.

***Mousley, A.** (2026, June). White matter connectivity associated with genetic risk distinguishes preadolescents' resilience to childhood adversity. The Child Development Forum, Centre for Child, Adolescent and Family Research, University of Cambridge, Cambridge, UK.

***Mousley, A.** (2026, March). The four phases of brain rewiring. Better Futures Programme, Professional and Continuing Education, University of Cambridge, Cambridge, UK.

Mousley, A. (2026, January). Topological turning points across the human lifespan. WLTS at MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK.

***Mousley, A.** (2025, September). Formation, refinement, and reorganisation of structural brain topology. University of Vermont, VT, USA.

***Mousley, A.** (2025, September). Topological turning points across the human lifespan. Flash Talk at FLUX, Dublin, Ireland.

***Mousley, A.** (2025, June). Topological turning points across the human lifespan. Power-Pitch at the Cambridge Imaging Festival, University of Cambridge, Cambridge, UK.

Mousley, A. (2023, October). Structural topology and network simulations across the lifespan. WLTS at MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK.

***Mousley, A.** (2023, September). Premature birth changes wiring constraints in neonatal structural brain networks. Centre for the Developing Brain at King's College London, London, UK.

Mousley, A. (2022, December). Quantifying and modelling brain organization. Methods Day at MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK.

***Mousley, A.** (2022, November). Simulating brain development. CamBrain Neurotalks, University of Cambridge, Cambridge, UK.

***Mousley, A.** (2022, September). A generative network model of neonatal brain development. Data Blitz at the Cambridge Neuroscience Seminar, University of Cambridge, Cambridge, UK.

* Indicates invited or selected to talk

WORKSHOPS & POLICY DEVELOPMENT

Invited attendee at CSaP roundtable on reframing mental health in policy making (2026, April). Hosted by the Centre for Science and Policy and the Downing Battcock Institute, Downing College, Cambridge, UK.

Invited to contribute to an evidence brief for the [Inquiry into the situation of young people not in employment, education or training \(NEET\)](#). All-Party Parliamentary Group on Poverty and Inequality, Remote.

Invited attendee at Diverse Trajectories to Good Developmental Outcomes Workshop (2022, November). *Global Scientific Conference on Human Flourishing*, Cambridge, UK.

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SCHOLARSHIPS & GRANTS

2021 – 2025	Gates Cambridge Scholarship, <i>Gates Cambridge Trust</i>	Approx. £201,500
2020	Katharine Jones Baker Fellowship, <i>Vassar College</i>	\$3,000
2019	Internship Grant Fund, <i>Vassar College</i>	\$2,500
2018	Internship Grant Fund, <i>Vassar College</i>	\$1,500

RESEARCH EXPERIENCE

May 2025 – Present	Post-Doctoral Research Associate	University of Cambridge, Cambridge, UK
• Gather, process, and organise diffusion weighted imaging datasets (9,000+ participants) using DSI Studio and Python. • Implement computational methods (e.g., weighted generative network modelling) to explore the relationships among childhood adversity, genetics, and brain structure. • Communicate research in conference presentations (poster and oral), peer-reviewed publications, and to the public in media (e.g., TV, radio and print articles). • Interview, supervise and train PhD and master's students. • Collaborate with researchers across departments at the University of Cambridge and international universities.		
Oct. 2021 – April 2025	PhD Candidate	University of Cambridge, Cambridge, UK
• Conducted computational neuroscience research using advanced scientific methods (e.g., diffusion tensor imaging fibre tracking, machine learning and generative network modelling) to deepen understandings of the development of diverse brain networks. • Identified appropriate datasets, completed data sharing and administrative duties to gather, organise, clean and preprocess data and published the data on OSF. • Implemented neuroimaging pipelines (e.g. QSIprep) and conducted analyses in MATLAB, Python and R and published code on GitHub. • Presented original research (e.g., presentations, posters) to audiences of experts and laypeople. • Wrote and revised peer-reviewed manuscripts with collaborators and supervisors.		
Aug. 2020 – Aug. 2021	Volunteer Researcher Assistant	University of Cambridge, Cambridge, UK
• Explored the neurobiological mediators/moderators of the decline of fluid intelligence in healthy, ageing individuals using a large, population-representative, lifespan cohort of fMRI data (CamCAN). • Developed MATLAB pipeline to investigate correlations between age, fluid intelligence and BOLD activation patterns in the multiple demands network while considering potential confounds (e.g., diet, education).		
Aug. 2017 – May 2020	Research Assistant	Vassar College, New York, USA
• Analysed mammalian behaviours using Smart Video Tracking software to quantify freezing, rearing, and strength attend behaviours and tested a priori predictions with SPSS, Excel, and JASP. • Sliced brains using the vibratome, cleanly mounted brain slices for histological processing, digitally reconstructed neuronal microstructures and analysed results of immunohistochemistry.		
May 2019 – Aug. 2019	Research Intern	École Polytechnique, Montreal, Canada
• Created optimised image processing pipelines to convert confocal microscopy images into ideal images for publication by integrating ImageJ, MATLAB and Python scripts to validate stainless microscopy. • Evaluated unsupervised machine learning methods (e.g., PCA, ICA and K-Means Clustering) for visualisation of myelin in the spinal cord.		

ACADEMIC & PUBLIC SCIENCE COMMUNICATION

June 2025	Blog: Embracing uncertainty in neuroscience
June 2025	Media interview with Nature
April 2026	Media interview with the Hindustan Times
Feb. 2026	Media interview with Vassar College
Feb. 2026	Media interview with Science Illustrated, Scandinavia
Jan. 2026	Media interview with Science & Vie
Jan. 2026	Media interview with PsyPost.org
Dec. 2025	Media interview with The Wall Street Journal
Dec. 2025	Live radio interview on CBC The Current
Dec. 2025	Podcast interview on Fun Kids Science Weekly
Nov. 2025	Live TV Interview on BBC News
Nov. 2025	Live TV Interview on Sky News
Nov. 2025	Media interview with NewScientist.com
Nov. 2025	Media interview with The Times

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Nov. 2025 [Media interview with The Sunday Times](#)
Nov. 2025 [Media interview with The Guardian](#)
Nov. 2025 [Media interview with BBC News](#)
Nov. 2025 [Media interview with The Washington Post](#)
Nov. 2025 [Media interview with Scientific American](#)
Nov. 2025 [Media interview with NBC News](#)
Nov. 2025 [Media interview with The Daily Mail](#)
Nov. 2025 [Media interview with National Geographic](#)
Nov. 2025 [Media interview with Sciences et Avenir](#)
Nov. 2025 [Media interview with Le Nouvel Obs](#)
Nov. 2025 [Media interview with EverydayHealth.com](#)
Nov. 2025 [Media interview with El Mercurio](#)
Nov. 2025 [Media interview with Der Standard](#)
Nov. 2025 [Media interview with Trouw](#)
Nov. 2025 [Media interview with El País](#)
Nov. 2025 [Live radio interview on BBC Radio 4 World at One](#)
Nov. 2025 [Radio interview on BBC Wales](#)
Nov. 2025 [Live radio interview on BBC World Service News Hour](#)
Nov. 2025 [Live radio interview on BBC Radio 5](#)
Nov. 2025 [Live radio interview on ABC Radio Perth Drive](#)
Nov. 2025 [Live radio interview on ABC North Coast](#)
Nov. 2025 [Live radio interview on ABC News Radio](#)
Nov. 2025 [Live radio interview on BBC Radio 5 Live Drive](#)
Nov. 2025 [Live radio interview on The Agenda on Dubai Eye 103.8](#)
Nov. 2025 [Radio interview on CBC Radio 1](#)
Nov. 2025 [Live radio interview on BBC Hereford & Worcester](#)

Oct. 2025 – Present **Postdoc Committee Member** MRC Cognition and Brain Sciences Unit, University of Cambridge
• Organise, run and host events, such as the all-day Postdoc Retreat, social lunches, and game nights.

Oct. 2025 **Navigating neurodiversity in the workplace** University of Cambridge
• Panellists for an event discussing building spaces where everyone can thrive.

Oct. 2025 **Big Biology Day Volunteer** Hills Road Sixth Form College, Cambridge, UK
• Ran a family-friendly neuroscience stall, answering questions from parents and children about the brain and ran hands-on activities.

Oct. 2023 – May 2024 **Event Host & Organizer** Pint of Science, Cambridge, UK
• Collaborated with a team to organise events (e.g., found speakers, coordinated venues, etc) for a worldwide science festival to share discoveries with the public.
• Hosted three events to communicate scientific concepts with laypeople, directing the audience and speakers, and encouraging engagement with local artists.

Guest author, Grey Matters (<https://www.greymattersjournalvc.org/issue-3>) Vassar College, New York, USA

Mousley, A. L. (2022, November). Simulating brains: how it works and what we learn. Presented to the public at the *Cambridge Festival*, Cambridge, UK.

CLINICAL, TEACHING, & MENTORING EXPERIENCE

April 2026 – Present **Mentor for the BNA Scholars Programme** British Neuroscience Association, Remote
• Provide professional and personal mentorship for graduate students in the UK, with particular focus on issues of diversity, equity and inclusion.

Oct. 2025 – Present **Co-supervisor** MRC Cognition and Brain Sciences Unit, University of Cambridge
• Supervise MPhil student as they design, conduct, and write up their master's thesis.

June 2023 – Present **Mentor** Lumiere Education, Remote
• Create educational programmes and syllabi tailored to individual mentees and guide weekly one-on-one work sessions to teach research skills and provide feedback.

Aug. 2019 – May 2020 **One-to-One Classroom Support** Wimpfheimer Nursery School, New York, USA
• Supported a five-year-old child with Down syndrome during an after-school programme.

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June 2018 – Aug. 2018

Clinical Intern

Chandelier Assessments, Vermont, USA

- Observed and aided in children's psychological assessments, including the Wechsler Adult Intelligence Scale (WAIS).

Jan. 2018 – May 2018

Teachers Assistant

Wimpfheimer Nursery School, New York, USA

- Aided teacher in a classroom of five and eight two-year-olds and developed and ran an educational programme.

HONOURS & SOCIETIES

Sigma Xi Scientific Research Honor Society (2020); Vassar College Honours (2020); Neuroscience and Behavior Departmental Honours (2020); Psi Chi International Honor Society in Psychology (2019); Liberty League All-Academic Team (2017-2019)

METHODOLOGICAL SKILLS

Languages: Python, R, MATLAB, Bash

Software and Tools: GitHub, Conda, Slurm, DSI Studio, FreeSurfer, QSIprep, Scikit-learn, Brain Connectivity Toolbox, TensorFlow, Open Science Framework (OSF)

Statistics and analyses: Tractography, Graph theory, Generative network modelling (GNM), Uniform manifold approximation & projection (UMAP), Factor analysis, Principal components analysis (PCA), Least absolute shrinkage and selection operator (LASSO), Generalised additive models (GAM), Linear mixed effects models (LMEM), Partial least squares regressions (PLS), Dynamic time warping (DTW), Morphometric INverse Divergence (MIND)

IDE: Visual Studio Code, Positron, JupyterLab, Jupyter Notebook, RStudio, Atom

Course Certificates: Advanced Learning Algorithms, Supervised Machine Learning: Regression and Classification, Unsupervised Learning, Recommenders, Reinforcement Learning